Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
AI(Artificial Intelligence)

Practical VII Task

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Practical 7 Part 2 & 3 Tasks

Q.1

Area (sq.ft.) Price (Lakhs)

Area (sq.ft.)	Price (Lakhs)
600	32
800	40
1000	49
1200	58
1400	68
1600	79

Find the regression equation.
Predict the price of a 1500 sq.ft. house.
Predict the price of a 2500 sq.ft. house.

Code:-

```
import pandas as pd
from sklearn.linear_model import LinearRegression

# Create dataset
data = {
   "Area":[600,800,1000,1200,1400,1600],
   "Price":[32,40,49,58,68,79]
}
df = pd.DataFrame(data)

X = df[["Area"]]
y = df["Price"]
# Train model
model = LinearRegression()
model.fit(X, y)
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

```
# Prediction for 1500 sq.ft.
price1500 = model.predict([[1500]])
# Prediction for 2500 sq.ft.
price2500 = model.predict([[2500]])
print("Predicted price for 1500 sq.ft:", round(price1500[0],2),"Lakhs")
print("Predicted price for 2500 sq.ft:", round(price2500[0],2),"Lakhs")
```

Output:-

```
Slope: 0.04685714285714286
Intercept: 2.790476190476191
Predicted price for 1500 sq.ft: 73.08 Lakhs
Predicted price for 2500 sq.ft: 119.93 Lakhs
/usr/local/lib/python3.12/dist-packages/sklearn/utils/vali
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/vali
  warnings.warn(
```

Q.2 A company has collected the following employee data:
• Employee ID
• Employee Age
• Years of Experience
• Salary
You need to develop a Linear Regression model to predict an employee's salary.
Tasks
1. Identify the independent variable (X) and the dependent variable (Y).
2. Explain why you selected these variables.
3. Create a Pandas DataFrame using the given data.
4. Split the dataset into training and testing sets.
5. Train a Linear Regression model.
6. Predict the salary for an employee having 7 years of experience.
7. Display:
o Slope (Coefficient)
o Intercept
o R2 Score
8. Plot the regression line.

Code:-

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
# Create Employee Dataset
data = {
    "Employee_ID": [101, 102, 103, 104, 105, 106, 107, 108, 109, 110],
    "Employee_Age": [22, 25, 28, 30, 35, 40, 32, 29, 45, 38],
    "Years_of_Experience": [1, 2, 3, 5, 8, 10, 6, 4, 15, 12],
    "Salary": [25000, 30000, 35000, 45000, 60000, 70000, 52000, 40000, 90000, 80000]
}
```

```
# Create DataFrame
df = pd.DataFrame(data)
print("Employee Dataset:\n")
print(df)
# Independent Variable (X)
X = df[["Years_of_Experience"]]
# Dependent Variable (Y)
y = df["Salary"]
# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=1
)
# Train Linear Regression Model
model = LinearRegression()
model.fit(X_train, y_train)
# Predict Salary for 7 Years Experience
prediction = model.predict([[7]])
print("\nPredicted Salary for 7 Years Experience: ₹", round(prediction[0], 2))
# Display Slope, Intercept and R² Score
print("\nSlope (Coefficient):", round(model.coef_[0], 2))
print("Intercept:", round(model.intercept_, 2))
print("R² Score:", round(model.score(X_test, y_test), 4))
# Plot Regression Line
plt.scatter(X, y, color="blue", label="Actual Data")
plt.plot(X, model.predict(X), color="red", label="Regression Line")

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Employee Salary Prediction using Linear Regression")
plt.legend()
plt.show()
```

Output:-

```
Employee Dataset:

   Employee_ID  Employee_Age  Years_of_Experience  Salary
0          101            22                    1   25000
1          102            25                    2   30000
2          103            28                    3   35000
3          104            30                    5   45000
4          105            35                    8   60000
5          106            40                   10   70000
6          107            32                    6   52000
7          108            29                    4   40000
8          109            45                   15   90000
9          110            38                   12   80000

Predicted Salary for 7 Years Experience: ₹ 54117.65

Slope (Coefficient): 4705.88
Intercept: 21176.47
R² Score: 0.9881
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: U
  warnings.warn(
```

Employee Salary Prediction using Linear Regression
Actual Data
Regression Line
Salary
90000
80000
70000
60000
50000
40000
30000
2
4
6
8
10
12
14
Years of Experience